Business Analyst Case Study: E-Commerce Return & Refund Management System

Role: Business Analyst | **Domain:** E-Commerce / Retail | **Deliverables:** BA Documentation + Working Prototype

1. Project Overview

Client: ShopEasy (fictional mid-size online retailer selling fashion, electronics, and home goods)

Problem statement: ShopEasy handles returns through email and phone. Customers cannot raise or track a return online, and the support team processes requests manually in spreadsheets. Return resolution takes 10-14 days, refund errors are common, and customer complaints about returns are rising.

Project goal: Design and prototype a self-service Return & Refund Management System where customers raise and track returns, and admins review and resolve them (approve, reject, refund, replace) with clear rules and an audit trail.

Business objectives (SMART)

#	Objective	Target
1	Reduce average return resolution time	From 12 days to 4 days within 6 months
2	Cut manual support effort on returns	Reduce return-related tickets by 40%
3	Improve customer satisfaction on returns	CSAT from 3.2 to 4.0
4	Reduce refund errors	Below 1% of processed returns

Scope

- In scope: Customer login/dashboard, My Orders, return request, return tracking, admin dashboard, approve/reject, refund, replacement, email notifications.
- Out of scope: Payment gateway development (refund is simulated), courier integration (status updated manually), warehouse inventory system, mobile app.

Assumptions: Customers are registered and logged in; orders exist in the system; refunds go back to the original payment method.

Constraints: Prototype built with a limited timeline and no live integrations.

2. Stakeholders

Stakeholder	Role	Interest	Influence	Engagement
Customer	End user	Easy, fast, transparent returns	High (drives revenue)	Keep satisfied, gather feedback
Customer Support Agent / Admin	Primary system user	Less manual work, clear decision rules	High	Key participant in requirements and UAT
Operations / Warehouse Manager	Receives returned goods	Know what is coming back and when	Medium	Consult
Finance Team	Processes refunds	Accurate refunds, audit trail	High	Consult and approve refund rules
Product Owner	Owns backlog	Delivers value on time	High	Manage closely
Legal / Compliance	Return policy adherence	Consumer-law compliance	Medium	Inform and review policy
Development and QA Team	Build and test	Clear, testable requirements	Medium	Collaborate daily
Sponsor (Head of E-commerce)	Funds project	ROI, reduced cost	High	Manage closely

3. As-Is Process

1. Customer emails or calls support with order number and reason.
2. Agent looks up the order in the order system.
3. Agent manually checks the return policy (window, condition, category).
4. Agent replies by email approving or rejecting; may need several back-and-forth emails for photos or details.
5. Customer ships the item; agent has no visibility of receipt.
6. Warehouse informs agent by email when item arrives.
7. Agent asks Finance to issue a refund, or Ops to send a replacement.
8. Agent emails the customer the outcome.

Pain Points

ID	Pain point	Impact	Root cause
P1	No way to raise or track a return online	Repeated "where is my refund?" tickets	No self-service
P2	Policy applied inconsistently across agents	Customer disputes, unfair outcomes	Rules live in people's heads
P3	Manual data entry in spreadsheets	Errors, wrong refund amounts	No central system
P4	No status visibility for customer or warehouse	Delays, duplicate follow-ups	Email-based communication
P5	Long back-and-forth for missing info	10-14 day resolution	No structured request form
P6	No audit trail or reporting	Cannot spot fraud or top return reasons	No data capture

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4. To-Be Process

1. Customer logs in, opens **My Orders**, selects an eligible order and item(s).
2. System checks eligibility automatically (return window, category, order status).
3. Customer submits a **Return Request** (reason, type: refund or replacement, comments, optional photo).
4. System creates a Return ID, sets status **Submitted**, and sends confirmation.
5. Admin reviews the request on the **Admin Dashboard** and **Approves** or **Rejects** (rejection reason is mandatory).
6. If approved, the customer is notified and status becomes **Approved - Awaiting Item**.
7. Warehouse/admin marks the item **Received and Inspected**.
8. Admin completes the outcome: **Refund** (status: Refund Processed) or **Replacement** (status: Replacement Shipped).
9. Request is **Closed**; the customer sees the full timeline on **Tracking**.

Status Lifecycle

```
Submitted -> Under Review -> Approved / Rejected -> Item Received -> Refund Processed / Replacement Shipped -> Closed
```

(Customer may **Cancel** only while status is Submitted or Under Review.)

Gap Analysis

Area	As-Is	To-Be
Request channel	Email/phone	Online form
Policy check	Manual	Automated eligibility rules
Tracking	None	Real-time status timeline
Decision	Ad hoc	Rule-based with mandatory reasons
Audit	None	Full status/action history

5. Business Requirements Document (BRD) Summary

BRD sections: Purpose, Business Objectives, Scope, Stakeholders, As-Is/To-Be, Functional Requirements, Non-Functional Requirements, Business Rules, Assumptions, Constraints, Risks, Success Metrics, Sign-off.

Risks

Risk	Likelihood	Impact	Mitigation
Fraudulent or repeated returns	Medium	High	Return-frequency flag, mandatory reasons, admin review
Policy changes mid-project	Medium	Medium	Keep rules configurable, change control process
Low adoption by support team	Low	High	Involve agents in UAT, provide training
Refund mismatch with payment provider	Low	High	Finance sign-off, reconciliation report

Success Metrics

Average resolution time, % of returns raised online, refund error rate, return-related ticket volume, CSAT.

6. Requirements

6.1 Functional Requirements

ID	Requirement	Priority (MoSCoW)
FR-01	Customer can log in and view a personal dashboard with order and return summary	Must
FR-02	Customer can view all their orders with status, date, items, and amount	Must
FR-03	System shows a "Return" option only for eligible items	Must
FR-04	Customer can submit a return request selecting item, reason, and resolution type (refund/replacement)	Must
FR-05	System generates a unique Return ID and sends confirmation	Must
FR-06	Customer can track return status with a timeline	Must
FR-07	Customer can cancel a return before it is approved	Should
FR-08	Admin can view all requests with filters (status, date, category) and search by Return ID/Order ID	Must
FR-09	Admin can approve a request	Must
FR-10	Admin can reject a request with a mandatory reason	Must
FR-11	Admin can mark an item as received	Must
FR-12	Admin can process a refund (full or partial with reason)	Must
FR-13	Admin can issue a replacement	Must
FR-14	System sends email/notification on each status change	Should
FR-15	Customer can upload a photo for damaged or wrong items	Could
FR-16	Admin dashboard shows KPIs (open requests, avg resolution time, top return reasons)	Should
FR-17	System maintains an audit log of all status changes and actions	Must

6.2 Non-Functional Requirements

ID	Requirement
NFR-01	**Performance:** Pages load within 3 seconds under normal load
NFR-02	**Security:** Role-based access; customers see only their own data; admin actions require login
NFR-03	**Usability:** Return request completed in under 2 minutes, responsive on mobile browsers
NFR-04	**Availability:** 99.5% uptime

ID	Requirement
NFR-05	**Auditability:** All decisions stored with user, timestamp, and reason
NFR-06	**Data privacy:** Personal data handled per applicable data protection rules

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7. Business Rules

ID	Rule
BR-01	Returns are allowed only within **30 days** of delivery date
BR-02	Only orders with status **Delivered** are eligible
BR-03	Non-returnable categories: innerwear, perishables, gift cards, customized items
BR-04	Reason "Damaged" or "Wrong item" requires a comment (photo recommended)
BR-05	Rejection requires a mandatory reason visible to the customer
BR-06	Refund goes to the original payment method; timeline 5-7 business days after item inspection
BR-07	Refund amount = item price (plus shipping fee only if return is due to seller error); no refund if item fails inspection
BR-08	Replacement is allowed only if the item is in stock; otherwise refund is offered
BR-09	One active return request per order item
BR-10	Customer may cancel only before approval
BR-11	Customers with more than 3 returns in 60 days are flagged for manual review
BR-12	Refund above a set threshold (e.g., INR 20,000) needs a second approver

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8. User Stories and Acceptance Criteria

US-01: View my orders (FR-02)

As a customer, **I want** to see all my orders and their status, **so that** I can choose which one to return.

Acceptance criteria

- Given I am logged in, when I open My Orders, then I see my orders sorted by latest first with status, date, and total.
- Given an order is not mine, then I cannot view it.
- Given I have no orders, then I see a friendly "No orders yet" message.

US-02: Raise a return request (FR-03, FR-04, FR-05)

As a customer, **I want** to submit a return with a reason, **so that** I can get a refund or replacement.

Acceptance criteria

- Given a delivered order within 30 days, then the "Return" button is enabled.
- Given an order past 30 days or a non-returnable item, then the button is disabled with an explanatory message.
- Given I submit without a reason, then the system shows a validation error.
- Given I submit successfully, then a Return ID is generated, status is "Submitted," and a confirmation is shown.
- Given an item already has an active return, then I cannot raise a second one.

US-03: Track my return (FR-06)

As a customer, **I want** to see the current status and history of my return, **so that** I do not need to contact support.

Acceptance criteria

- Given I open a return, then I see the current status and a timeline with dates.
- Given a rejection, then I see the rejection reason.
- Given a refund is processed, then I see the amount and expected credit time.

US-04: Cancel a return (FR-07)

As a customer, **I want** to cancel my request if I change my mind.

Acceptance criteria

- Given status is Submitted or Under Review, then a Cancel button is available.
- Given status is Approved or later, then Cancel is not available.

US-05: Review requests (FR-08)

As an admin, **I want** to see all requests with filters, **so that** I can prioritize my work.

Acceptance criteria

- Given I open the dashboard, then I see requests with Return ID, customer, item, reason, date, status.
- Given I apply a filter or search, then the list updates accordingly.

US-06: Approve or reject (FR-09, FR-10)

As an admin, **I want** to approve or reject a request, **so that** valid returns proceed and invalid ones are closed with a reason.

Acceptance criteria

- Given a request under review, when I approve, then status becomes "Approved" and the customer is notified.
- Given I reject without a reason, then the system blocks the action.
- Given I reject with a reason, then status becomes "Rejected" and the reason is shown to the customer.
- Given any action, then it is recorded in the audit log.

US-07: Process a refund (FR-11, FR-12)

As an admin, **I want** to issue a refund after the item is received, **so that** the customer is paid correctly.

Acceptance criteria

- Given the item is marked "Received," then the Refund action becomes available.
- Given I enter a partial amount, then a reason is required and the amount cannot exceed the item price.
- Given the refund is processed, then status becomes "Refund Processed" and the customer is notified.
- Given amount exceeds the threshold, then a second approval is required (BR-12).

US-08: Issue a replacement (FR-13)

As an admin, **I want** to send a replacement, **so that** the customer receives the right item.

Acceptance criteria

- Given stock is available, then I can create a replacement and status becomes "Replacement Shipped."
- Given stock is unavailable, then the system prompts me to offer a refund instead.

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9. Use Cases

UC-01: Submit Return Request

Field	Detail
Actor	Customer
Precondition	Customer is logged in; order is delivered and within return window
Trigger	Customer clicks "Return" on an order item
Main flow	1. Select item. 2. Choose reason. 3. Choose refund or replacement. 4. Add comments/photo. 5. Submit. 6. System validates eligibility. 7. System creates Return ID and confirms.
Alternate flow	A1: Item ineligible, system blocks and shows the reason. A2: Missing mandatory fields, system shows errors.

Field	Detail
Exception	E1: System error, request not saved; user is asked to retry.
Postcondition	Return created with status "Submitted"; notification sent
Business rules	BR-01 to BR-04, BR-09

UC-02: Review and Decide on Return

Field	Detail
Actor	Admin
Precondition	Return exists with status "Submitted" or "Under Review"
Main flow	1. Admin opens request. 2. Reviews details and order history. 3. Approves. 4. System updates status and notifies customer.
Alternate flow	A1: Admin rejects, must enter reason. A2: Customer flagged (BR-11), admin escalates for manual review.
Postcondition	Status is Approved or Rejected; audit entry created

UC-03: Process Refund or Replacement

Field	Detail
Actor	Admin (Finance for high-value approval)
Precondition	Item received and inspected
Main flow	1. Admin selects Refund or Replacement. 2. System calculates refund amount. 3. Admin confirms. 4. System updates status, notifies customer, closes the request.
Alternate flow	A1: Inspection failed, refund rejected with reason. A2: Out of stock, refund offered. A3: Amount above threshold, second approver required.
Postcondition	Status is Refund Processed or Replacement Shipped

UC-04: Track Return

Actor: Customer. Opens My Returns, selects a Return ID, views status timeline and any admin messages.

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10. Test Scenarios

ID	Scenario	Type	Expected result	Req
TC-01	Return an item delivered 10 days ago	Positive	Request created, Return ID generated	FR-04, BR-01
TC-02	Return an item delivered 31 days ago	Negative	Return option disabled with message	BR-01
TC-03	Return an order still "Shipped"	Negative	Not eligible	BR-02
TC-04	Return a non-returnable category item	Negative	Blocked with message	BR-03
TC-05	Submit form without reason	Negative	Validation error	FR-04
TC-06	Submit second request for same item	Negative	Blocked (duplicate)	BR-09
TC-07	Customer views another customer's return	Security	Access denied	NFR-02
TC-08	Cancel return while Submitted	Positive	Status changes to Cancelled	BR-10
TC-09	Cancel return after Approved	Negative	Cancel not available	BR-10
TC-10	Admin approves request	Positive	Status Approved, notification sent	FR-09
TC-11	Admin rejects without reason	Negative	Blocked	BR-05
TC-12	Admin rejects with reason	Positive	Customer sees reason	FR-10
TC-13	Refund before item received	Negative	Refund action unavailable	FR-11
TC-14	Partial refund exceeding item price	Negative	Validation error	BR-07

ID	Scenario	Type	Expected result	Req
TC-15	Full refund after receipt	Positive	Status Refund Processed	FR-12
TC-16	Replacement when out of stock	Negative	Refund suggested	BR-08
TC-17	Customer with 4 returns in 60 days	Edge	Flagged for manual review	BR-11
TC-18	Refund above threshold	Edge	Second approval required	BR-12
TC-19	Every status change	Audit	Log entry with user and timestamp	FR-17
TC-20	Page load with 1,000 records	Non-functional	Loads under 3 seconds	NFR-01

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11. Requirements Traceability Matrix (RTM)

Business Objective	Requirement	Business Rule	User Story	Use Case	Test Case	Status
Reduce tickets (Obj 2)	FR-02	-	US-01	-	TC-07	Built
Reduce resolution time (Obj 1)	FR-03, FR-04, FR-05	BR-01 to BR-04, BR-09	US-02	UC-01	TC-01 to TC-06	Built
Reduce tickets (Obj 2)	FR-06	-	US-03	UC-04	TC-19	Built
Improve CSAT (Obj 3)	FR-07	BR-10	US-04	-	TC-08, TC-09	Built
Reduce resolution time (Obj 1)	FR-08	-	US-05	UC-02	TC-20	Built
Reduce resolution time (Obj 1)	FR-09, FR-10	BR-05, BR-11	US-06	UC-02	TC-10 to TC-12, TC-17	Built
Reduce refund errors (Obj 4)	FR-11, FR-12	BR-06, BR-07, BR-12	US-07	UC-03	TC-13 to TC-15, TC-18	Built
Improve CSAT (Obj 3)	FR-13	BR-08	US-08	UC-03	TC-16	Built
Audit and control	FR-17	-	US-06, US-07	UC-02, UC-03	TC-19	Built

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12. Working Tool (Prototype)

Customer side

- Dashboard: summary of orders and active returns
- My Orders: list with eligible-return indicator
- Return Request: form with validation of BR-01 to BR-04
- Tracking: status timeline

Admin side

- Admin Dashboard: request list, filters, KPIs
- Approve / Reject (mandatory reason)
- Refund (full/partial) and Replacement actions
- Audit log

Suggested tech: HTML/CSS/JS front end with a simple backend (Node/Flask) and SQLite, or a no-code/low-code option. Include screenshots and wireframes in `/screenshots` and `/wireframes`.

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13. Suggested GitHub Repository Structure

```
returns-refund-ba-project/
|-- README.md                    (this case study summary)
|-- 01-stakeholders/
|-- 02-as-is-to-be/              (process flows, BPMN/swimlanes)
|-- 03-brd/
|-- 04-requirements/             (FR, NFR, business rules)
```

```
|-- 05-user-stories/
|-- 06-use-cases/
|-- 07-test-scenarios/
|-- 08-rtm/
|-- 09-wireframes/
`-- 10-working-tool/              (source code, screenshots, demo link/video)
```

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14. Outcomes (Projected)

Metric	Before	After (target)
Avg resolution time	12 days	4 days
Returns raised online	0%	90%
Return-related tickets	100% baseline	-40%
Refund error rate	~5%	<1%
Return CSAT	3.2	4.0

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15. Interview Talking Points

30-second pitch: "I analyzed a returns process handled by email and spreadsheets, documented the as-is and to-be flows, wrote a BRD with requirements, business rules, user stories and acceptance criteria, traced everything in an RTM, and built a working prototype so stakeholders could validate the requirements."

Questions to prepare for

1. *How did you prioritize requirements?* MoSCoW, tied to business objectives; core return flow and admin decisions were Musts.
2. *How did you handle conflicting stakeholders?* Support wanted flexibility, Finance wanted strict control; I proposed configurable rules plus the threshold-based second approval (BR-12).
3. *How did you ensure completeness?* RTM linking objectives to requirements to tests; walkthroughs with each stakeholder group.
4. *How did you write acceptance criteria?* Given/When/Then, one testable behavior each, covering positive, negative, and edge cases.
5. *What would you do about scope creep?* Change request process, impact analysis, backlog re-prioritization with the Product Owner.
6. *What was the hardest requirement?* Refund rules (partial refunds, inspection failure, thresholds) because they involve money and multiple teams.
7. *How do you measure success?* The KPIs in section 1 against baseline.

Tools to mention: Draw.io/Lucidchart (flows), Figma/Balsamiq (wireframes), Jira (backlog), Excel (RTM), Postman (API testing if applicable), GitHub (portfolio).

Note: ShopEasy is a fictional client created for portfolio purposes. Numbers are illustrative targets, not real data.